Technical Instructions

Research Accommodation

Feature Story

Magazine Layout

Infographic

Data Report

Text Revision and Analysis

Contract Example

RESEARCH ACCOMMODATION

PROFESSIONAL AND TECHNICAL WRITING

PROJECT TITLE

Out of Sight, Out of Mind?

DESCRIPTION

I designed this project to function as an accommodation piece to a research article. I examined research from the University of Austin concerning how a smartphone affects a user's cognition and memory then wrote the piece to inform a wider audience of the results.

AUDIENCE

General Audience

SKILLS DEVELOPED

- Analyzed complex information and prepared it for a non-technical audience
- Created a clear and coherent title, billboard description, and lede to catch attention and direct the piece
- Defined complex terms and technical information for an intended audience

APPLICATIONS USED

- Microsoft Word
- Adobe InDesign

OUT OF SIGHT, OUT OF MIND?

Smartphones command our attention throughout the day, but could the devices tug at our minds even when not in use? A recent study suggests unseen effects on memory and cognition.

A sharp ping cuts through the air, or maybe a quick volt shakes the table. You look around the room and wonder whether you are the one summoned. For the moment you lose your place in the book you are reading, or maybe you pause mid-sentence while composing an email. Things settle back down, or at least they do until the next notification rings through. But just what happens until then? Is it possible that your smartphone doesn't need to actively signal to get your attention?

A new study from the University of Austin suggests that the mere presence of our smartphones can draw attention away from a task. Researchers found we feel this effect even when a phone is powered off or face down in front of the user.

Our Smartphones have a unique command over us. Previous surveys clock the average number of interactions a user has with their phone at 85 per day. For many the screen is the first light of the day and the last before bed. These changes all come in a relatively short timeframe. In 2007, 7% of American adults owned a smartphone. One decade later, ownership is now around 77%, with 92% of adults under 35 equipped with one. The phones are almost literally attached to the hip on most people, just waiting to be picked up and used to entertain, inform, or possibly distract. But as the study suggests, a smartphone could do the latter without leaving its place on the desk.

Researchers set up an experiment where the main variable was set as the location of a user's smartphone. Over 500 undergraduates participated in total. Researchers randomly assigned the students to one of three phone location conditions. They were to place their phone either on the desk within sight, in their pocket or bag, or outside of the testing room completely. Students were left unaware of the researchers' real intent as they were then administered a cognitive test.

The test, known as an Operational Span, is widely used to evaluate several cognitive abilities, including both working memory and fluid intelligence. "Working Memory" refers to the theoretical system that selects, maintains, and processes information relevant to current tasks, and "fluid intelligence" represents the ability to reason and solve problems and goals. The test measured both by having students memorize a set of words while simultaneously solving math equations. It's a perfect litmus test that strains the limited pool of resources that make up the memory system. Any stimuli can leech resources from the mind, and the question was whether the phones could do so without being in use.

Results gave a resounding yes. Student's working memory capacity and fluid intelligence scores were noticeably different between the three groups. Participants who placed their phones on the desk in front of them displayed the lowest cognitive capacity, even when the phones were powered off. Students who had their phones tucked away in either their pocket or bookbag followed closely behind. Test scores were consistently highest with students whose phones were left outside the room completely. The results were clear – a student's memory capacity decreased as his or her phone was placed closer.

After the tests were completed, students filled out a questionnaire that asked about their experience. Researchers sought to get a better idea of the relationship between a student and their phone and whether they believed the device could affect their performance. When asked if the location of the phone affected their score, 75% of participants answered "not at all" and 85% said that it "neither helped nor hurt". The contrast between these answers and the results showed that few students could conceive that their phone could pull attention while not in use. The surveys also determined that students who reported being very dependent were most susceptible to distraction. Participants who admitted being heavily attached to their phone scored 20% lower than the other groups on the OSpan.

With such a massive change in technology in a short time period, the study offers an important check on how we interact with our smartphones. Devices grow faster by the day and add more ways to link to the world. Their pull on our finite attentional resources will continue to grow in the future. Soon we will have a whole generation that was equipped with rapid speed connectivity in their formative years. The students' test results prove that we don't yet understand the full scope of how these devices affect us. More research is needed to understand what other consequences lurk behind our screens.

The study has generated many more questions about our technology dependence, and debates will rage on between advocates and detractors. One thing that can be agreed upon is that smartphones will still dominate our daily lives. Only the most extreme luddite would argue these devices bring no improvements to society. Both the free exchange of information and connectivity create opportunities for people daily. However, these benefits cannot be taken for granted. Research like the Austin study proves that our relationship to technology can be a doubled edge sword, and it raises a caution as we race towards the next shiny new version of the device by our side. ■